

Project 2: Identifying Assets and Activities to Be Protected

Cole Dombrowski - MISY 5325

This report identifies the resources and business functions that support the Sabine Coastal Health Network (SCHN). It defines the scope and value of these assets to help align security with organizational priorities. The report also considers SCHN's new telehealth and remote patient monitoring initiatives and their impact on the organization's security needs.

1. Project 1 Review and Scenario Change

The new tele-health and remote patient monitoring program introduces a third-party cloud-hosted architecture and extends patient data access onto unmanaged home environments.

Assumptions

- Health data was assumed to stay on managed clinic terminals and was expanded to include personal computers/mobile apps using public networks

Scope and Boundaries

- Scope expands beyond SCHN facilities/internal networks
- SaaS gateway, external APIs, and home networks

Stakeholders: Adds the SaaS provider's tech support team, remote patients and clinical teams

External Dependencies: SCHN depends on the SaaS provider's services

Organizational Priorities

- Data availability and integrity become critical to patient safety
- Clinicians rely on telemetry data to identify emergencies outside clinics

2. Important Business Activities

1. Patient Care and Clinical Operations

- Emergency treatment, outpatient care, and diagnostics
- Critical to SCHN's healthcare mission
- Manual and automated processes

2. Electronic Health Records Management

- Collects, maintains, and distributes patient medical records
- Patient care support and clinical decision-making
- Automated system reliance

3. Telehealth and Remote Patient Monitoring

- Virtual care and collection of remote biometric data
- SCHN's care capabilities extends beyond its facilities
- Relies mainly on automated systems and external services

4. Billing, Payments, and Financial Administration

- Patient billing, payments, payroll, and financial records
- SCHN's financial operations are supported
- Automated system reliance

3. SCHN Asset Inventory (TABLE PAGE)

Asset or Asset Group	Category	Business Purpose	Primary Users/Owners	Important Dependencies
EHR Core Platform	Data & Information	Stores and manages patient and clinical records	Clinical and administrative staff	Network, workstations, database systems
Telehealth SaaS Cloud System	Cloud & Third-Party Services	Supports virtual care and remote patient monitoring data	Patients and clinical staff	Internet, vendor services, APIs
Patient Biometric Monitoring Units	Hardware & Medical Technology	Collects and transmits patient health measurements	Patients and clinical staff	Internet, SaaS platform, power
Core Corporate Network	Network & Communication	Connects systems and enables secure data communication	IT staff and organizational users	Firewalls, routers, WAN connections
Clinical Workstations & Point-of-Care Endpoints	Hardware & Medical Technology	Provides access to clinical and business systems	Clinical and administrative staff	Network, EHR, authentication
Financial & Billing Systems	Software & Information Systems	Processes billing, payments, and financial information	Finance and administrative staff	Network, databases, payment services
IT & Cybersecurity Personnel	Personnel	Maintains systems, security controls, and user access	IT and security teams	Network, systems, security tools
Five Core Operating Facilities	Facilities	Houses clinical operations, technology, and personnel	Patients, clinical, and administrative staff	Power, network, physical security

4. Activity-to-Asset Dependencies

Telehealth Services: Depends on the Telehealth SaaS Cloud System, internet connectivity, clinical workstations, authentication services, and clinical personnel

Remote Patient Monitoring: Depends on biometric monitoring units, the Telehealth SaaS Cloud System, integration APIs, internet connectivity, and the EHR platform

Patient Care Operations: Depends on physical facilities, clinical workstations, the EHR platform, medical personnel, and network access

Billing and Payments: Depends on financial and billing software, the corporate network, security controls, financial personnel, and external payment services

Key New Dependencies: Telehealth and remote monitoring introduce dependencies on patient home networks, internet connectivity, third-party cloud services, external APIs, and vendor availability

5. Five Priority Assets and Activities

1. Electronic Health Record Core Platform

Why: Contains clinical information for 70,000 active patients and supports patient care

Consequences: Loss or compromise could disrupt care, expose information, create regulatory issues, and damage SCHN's reputation

2. Telehealth SaaS Cloud System

Why: Provides an external platform for telehealth and remote monitoring

Consequences: An outage could prevent clinicians from accessing remote patient information and disrupt patient care

3. Telehealth and Remote Patient Monitoring

Why: Introduces patient-facing technology, home networks, and third-party systems into clinical operations

Consequences: Failures or inaccurate data could prevent clinicians from identifying patient emergencies and increase patient safety risks

4. Core Corporate Network

Why: Connects SCHN's facilities, systems, and users and supports data flow

Consequences: A network outage could disrupt clinical operations, access to patient records, communications, and other business activities

5. IT & Cybersecurity Support Group

Why: Responsible for maintaining SCHN's technology and security controls as it supports the expanded telehealth environment

Consequences: Limited capacity/human error could lead to security weaknesses

6. Management Brief

Primary Protection Targets

SCHN must prioritize protecting the EHR, clinical data, core networks, and systems that support patient care, as keeping patient information confidential, accurate, and available is important for patient safety and regulatory responsibilities.

Technology and External Dependencies

Telehealth and remote monitoring depend on third-party cloud services, internet connectivity, external APIs, and patient-owned devices and networks. Problems with these dependencies could disrupt remote care and access to patient information.

Impact of Telehealth Expansion

Telehealth expanded SCHN's technology boundary beyond its physical facilities. SCHN now has to consider unmanaged devices, home networks, cloud services, and other external dependencies when protecting clinical operations.

Next Phase

Management should use these priorities to focus limited resources on the assets/activities that have the greatest impact on patient safety and business operations. The next phase should evaluate the threats, vulnerabilities, and risks associated with these dependencies.

References:

- Gibson, D., & Igonor, A. (2020). *Managing Risk in Information Systems* (3rd ed.). Jones & Bartlett Learning.